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RESEARCH ARTICLE

A Novel Hybrid Approach to Forecasting Crude Oil Prices Using Local Mean Decomposition, ARIMA, and XGBoost

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ABSTRACT Crude oil in its usage and process brings about several issues that affect the environment including soiling of the air and water and the release of greenhouse gases leading to changes in the climate. Further, historically use of oil has created political and economic stakes that primarily end up in war, nationalism to control resources, and political instability. As a new approach to this problem in this study, we present a novel hybrid of Local Mean Decomposition (LMD) for breaking down the data and contribution coefficients for rebuilding the component with less computational complexity. Using the XGBOOST approach, deep dynamic learning features, augmented with autoregressive moving average (ARIMA), it is possible to predict each reconstructed segment. Turbulences that occur in the oil markets are random, the market shares are sensitive to pressures from other markets, the movement in the market is disorderly, and there is high volatility in the market. Thus, the price of oil is very unfixed, unpredictable, sensitive, and uncertain. Several techniques have been adopted in cost estimation in the aspect of time variation; some of the traditional methods include EMD, EEMD couldn't solve the computational complexity and save time and cost. Because of the unpredictability and fluctuation observed in oil prices non deep neural network models and data mining cannot be used in the prediction of costs and generalized forecasting. The analysis of the onset of deep non-linear modeling as a part of the machine learning process demonstrates how the discussed hybrid model successfully acquires relevant data and adapts a non-stationary function. The data is modeled using an LMD-SD-ARIMA-XGBOOST hybrid recurrent network when the nonlinearity is considered, and the residual values are forwarded to the XGBOOST model. Besides the volatility problem, the solution provided by the work also addresses the overfitting problem of neural networks. The hybrid LMD-SD-ARIMA-XGBOOST model overcomes overfitting by combining statistical and machine learning components, where the decomposition process isolates key features of the data, reducing noise and complexity. Unlike standalone neural networks, which require large amounts of data to generalize effectively, our hybrid approach leverages ARIMA and XGBOOST models that are inherently less prone to overfitting due to their simpler structures and feature-focused learning. To validate the proposed hybrid approach, data from the West Texas Intermediate (WTI) which is publicly available is employed.

INDEX TERMS Hybrid renewable energy system, safe working environment, energy efficiency, ARIMA, crude oil prices, LMD, stochastic and deterministic influences, XGBOOST.

I. INTRODUCTION

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Crude oil is among the key strategic resources of today's industrial relations with the social context. Some people

refer to it as the blood of the industry and it can assuredly not be considered as negligible as far as the prosperity and stability of any country is concerned. Moreover, this matter is considered the premise and foundation for developing civilization. [1]. Crude oil's nature in determining the security of nations, culture, and economic growth therefore places it at a vantage and strategic position in the present world, especially in the industrialized world. This has been done by using several factors, numbers of integrated forecasting models, econometric models, and artificial intelligence models for anticipation of the shift in the price of crude oil. This is so because crude oil prices are coined and contain a cracked surface and more to that, are nonlinear; the economic model thus, cannot predict such changes though at the same time as it can predict the regular steady and linear data. Therefore, one turns to such curved line models as random forests, artificial neural networks, support vector regression, and others. In a bid to eliminate some challenges that are linked to the absence of time homogeneity and instability, several scholars have used some measures. and the behavior of the series; in this manner, we understand how one can enhance the predicted accuracy of the CO price. These models of ARIMA-GARCH used in simulating and forecasting the conditional mean and volatility of the weekly price of crude oil have been done by Hassan and Su [2], where the above estimate and feasibility analysis have been carried out. Xie and associates [3]. Coupling the result of this with the previous study which focused on the monthly price data of WTI crude oil and the ARIMA model, a differential variable autoregressive model is employed from the decomposition of the monthly price data. Also, based on the concept of index Google effect of August 2011 the authors in [4] examined the feasibility of using the GIE for setting the price of CO and for its prediction using the ARIMA and ARMA-GARCH models. Here it is den comparing the previously used discrete and irregular models with the presented MMGARCH (mixture memory GARCH model). Klein and Walther [5] add to this body of knowledge by looking at the factors, that are linked to the price returns of CO stocks and building the puzzle pieces of a valuation model. Stelios and associates [6] The study will then present the comparison of the variances of the RW and AR models with the variances of the VAR model. This is shown in Table 1 above as the top list of results. In its integral conception, the economic model assumes that data is linear and follows a rather transparent, continuous pattern. In this way, it assists the economic model in forecasting the price of CO with fairly good accuracy. On the other hand, the crude oil prices on the international market have characteristics of being multidimensional, Sweden complex, and even possibly non-linear in nature. It is worthy of note that the attributes that have been accumulated in crude oil and hidden within the fluidity of the product may not be detectable by these often basic measurement procedures. Given that the development of AI is very rapid, conventional linear models such as SVR, RF, ANN, and others are used for the prediction of the CO price to obtain the non-linear CO price data set. For instance,

Chiroma et al. [7] reveal the price forecast of WTI CO with the help of the genetic algorithm and the neural network. Likewise, using a neural network, Baruník et al. [8] look at the crude oil futures price in an attempt to forecast the term structure of the same. Using (ICA-SVR) Imperialist Competitive Algorithms with Support Vector Regression, Fan, et al. [9] have attempted to forecast the price of crude oil. El-Masry and Mostafa [10] use the price of crude oil; and the stream learning by Gao and Yalin [11] for predicting the CO price. The literature review proved that the new model of artificial intelligence was better than the old model of artificial intelligence. This is why there are gaps in the time series of complex CO prices: The AI cannot be synthesized into a single ship. On the other hand are non-stationary, non-linear sequences that can be forecasted with good accuracy by an AI model. The hybrid forecasting model, therefore, minimizes the effect of non-stationary time series and non-linear characteristics of CO prices to enhance the precision of the several models in the estimation of CO prices. The integrated models have registered some relative improvement in the last few years to model and project the Co prices. Like Tao et al., [12] an EMD-SBM-FNN model was developed and in this context, it is visible that the developed model is comparatively more effective and has the perfect ability to capture the non-linearity of the price of the crude oil. Originally, the method named EEMD-PSO-based LSSVM-GRACH to predict primary and CO2 emission prices was instantiated by Zhang et al. in [13]. On this basis, Yu et al. [14] utilized the price of CO by the newly developed EEMD-DCD-LSSVR model. The authors of this paper cited from [15] and stated that SDAE and bagging can be used to predict the cost of CO. Following what has been done by the authors of [16], the authors sought to predict the price of CO using the EEMD-RVFL model. Moreover, the authors presented the EEMD-EELM-ADD model mentioned in [17] as a particular decomposition-ensemble model for forecasting the price of the CO. Ding [18] applies an integrated model known as EEMD-ANN-ADD to predict the price of CO. In so doing, the authors in [19] designed the DFN-AI model with the aspiration to make forecasts about the price of CO. However, the authors in [20] employed the VMDICA-ARIMA to model and, in an attempt to predict the price of CO. Zhang et al. following the same trend on the same note came up with a method of predicting the price of CO through an iterative combination [21]. RW and ARMA Have been employed in the study by the authors in [22] to forecast the price of CO. Concerning the prediction of the CO prices, the authors in [23] employed EEMD and the Dynamic Artificial Neural Network (DANN). The load forecasts used by the authors in [24] were included in the following models: Back Propagation Neural Network and Local Mean Decomposition- Back Propagation Neural Network-Long short-term memory (BPNN-LMD-LSTM). Hence, the design of the wind farm is in terms of micro-grids E-d by a planned time consistency along with the probability of random delay. The authors of [25] There was an introduction of a landslip displacement prediction model

based on the time-frequency analysis method: the previously proposed local mean decomposition-bidirectional long short-term memory, which is referred to as LMD-BiLSTM. The authors Heng Sun employ a three-step procedure that may have the potential to be applied in making RUL predictions in rotating Machines [26]. LMD and WTD have been added to a new model which is called LMD-WTD-LSTM because the writers of LSTM believed that it would forecast the short-term consumption of petrol in [27]. Similarly, in [28], Ya-Ni Lu put forward a certain model which in reality enhanced the prediction effect indeed. Utilizing the secondary factors, a new approach known as the variational mode decomposition (VMD) was employed to forecast the time series of the determinant factor. Such a forecast offers an accurate and rich expectation from a new model called the multiscale forecasting model. [29]. This model was able to complete the task of forecasting the difficult time series data than that of the other comparator model. VMD was used by the Same authors [30] to break down the given data into several features Authors in [30] used the VMD too; The multi-features are then used to train machine learning classifiers as shown as follows: The method adopted in this work is the Moving Window Ensemble Method, and therefore, Dynamic LMD (or RLMD) presented in [31] was used to forecast the daily quantities of PM10 and PM. Compared the nonlinear along with the linear modeling systems of five data. Talking about the research that had been done earlier the point to be noted is that the hybrid model is comprised of numerous different models and the strength that one model has will help overcome the shortcomings of the other models. Therefore, according to the research done above, it is preferred to the application of a single model because it utilizes many models to counter the strength of one model with the weakness of the other. The following are some of the questions that arise in this research; In what ways can the microprocessors make the CO price forecasts accurate? This brings the following question how the effect of the endpoint changes of the crude oil price time series can be eliminated? In this study, the authors predict the CO price with the assistance of LMD and an artificial intelligence model comprising the decomposition of the time series using LMD, using the AMI for effective computation of this LMD, modeling the volatility characteristics of the series with an econometric model, and finally, through the integration of all these LMDs using XGboost. In contrast to more conventional techniques like EMD and wavelet transformations, LMD is especially made to decompose a time series into intrinsic mode functions (IMFs) based on local averages, enabling a more flexible decomposition. LMD is highly effective in extracting pertinent information from noisy signals, while other techniques, such as EMD and EEMD, may suffer from mode mixing or require more computing power. The efficient computation of LMD through the use of AMI (Approximate Median Indicator) enhances computational efficiency, which may be essential when working with big datasets or real-time prediction jobs. Compared to EMD or wavelet transforms,

which may necessitate more complex computation or parameter adjustment, LMD in conjunction with AMI can provide a more effective and dependable denoising technique. The study is structured into four sections: These are the four parts through which every research study is divided which are, part one, 'Introduction', part two, 'methods', part three, 'analyses, and part four 'the conclusion. Section I therefore comprises the introduction and the literature review section. As will be seen in Section II, there are only a few words on the techniques used in the study. The results of the study are presented in Sect. III and the discussion of the results is provided in Sect. IV. Finally, the last section of the study is the conclusion which states a summary of the findings of the research besides the implication of the findings.

II. METHODOLOGY

A. LOCAL MEAN DECOMPOSITION (LMD)

LMD is a method that uses time-frequency analysis that is adaptive to handle non-stationary signals. The LMD method includes extracting the original signals from different envelope signals along with only signals with frequency modulation. The first signal, $x(t)$, is decomposed into the following five steps:

Local extremum points o_i are selected from $x(l)$, the initial signal, and then calculated by the average m_i of nearby extremum points o_i, o_{i+1} and the estimated envelope

$$\alpha_i = \frac{|o_i - o_{i+1}|}{2}, \quad (1)$$

- i. Using the local means m_i and the envelope estimate α_i to smoothed by using a moving average to $m_{11}(l)$ and $\alpha_{11}(l)$;
- ii. By ignoring The function of the local mean $m_{11}(l)$ in the initial signature $x(l)$:

$$h(l) = x(l) - m_{11}(l), \quad (2)$$

That is Divided by $\alpha_{11}(l), h_{11}(l)$ into the amplitude demodulated;

$$s_{11}(l) = h_{11}(l)/\alpha_{11}(l), \quad (3)$$

simply repeated Iterations of the frequency-modulated signal o times till $s_{1o}(l)$. Iterations are stopped when

$$\lim_{o \rightarrow \infty} \alpha_{1o}(l) = 1,$$

- iii. We can get an envelope $\alpha_1(l)$ and the initial component $PF_1(l)$ as:

$$\alpha_1(l) = \alpha_{11}(l)\alpha_{12}(l) \dots \alpha_{1o}(l) = \prod_{q=1}^o \alpha_{1q}(l), \quad (4)$$

$$PF_1(l) = a_1(l)s_{1o}(l), \quad (5)$$

- iv. The integrant $PF_1(l)$ is now taken out of $x(l)$ and the updated signal $u_1(l)$ is utilized to repeat the procedure k times to achieve a steady signal $u_k(l)$ and there are no

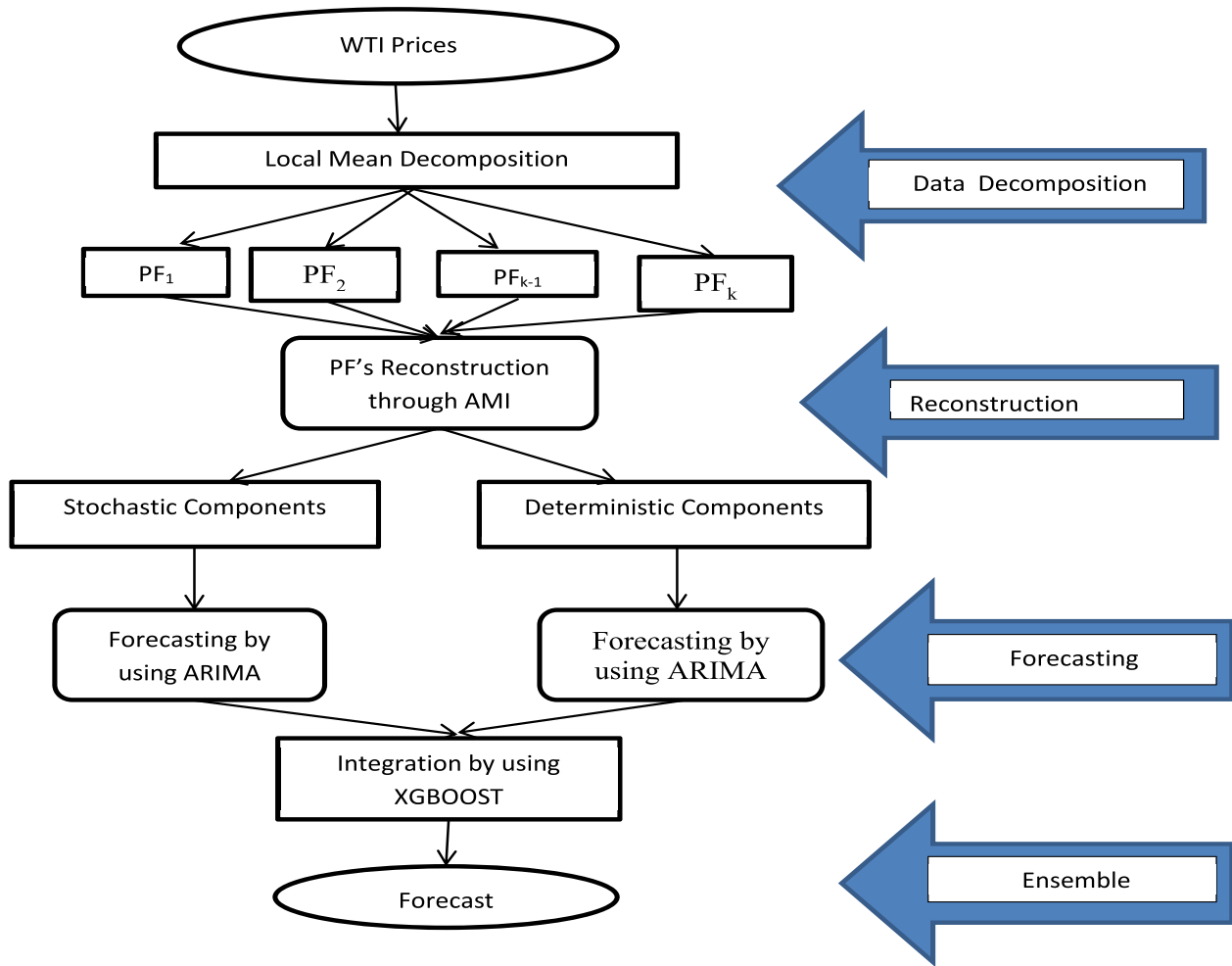


FIGURE 1. Flow chart of the proposed model.

longer any oscillations.,. Lastly, the initial signal $x(l)$ is defined as:

$$x(l) = \sum_{p=1}^k PF_p(l) + u_k(l), l = 1 \text{-----} k, \quad (6)$$

B. "AUTOREGRESSIVE INTEGRATED MOVING AVERAGE"

First, Around the 1970s, Box and Jenkins, introduced the ARIMA model. It consists of an ARIMA(p,d,q)-based autoregressive integrated moving average model. The dependent variable must be stationary for the model to work, which is accomplished by the I-component. The model also contains independent variables that are lags of the AR component (dependent variable) and/or error lags. The ARIMA model is essentially a regression model type with an MA part. The ARMA design is represented by equation

$$Y_l = \alpha_0 + \sum_{i=1}^p \alpha_i Y_{l-i} + \sum_{j=1}^q \beta_j \varepsilon_{l-j} + \varepsilon_l, \quad (7)$$

where p as well q denote the quantity of the terms MA and AR, respectively, and where l represents time and β and α denote the AR and MA coefficients, respectively. The

observations must be checked for stationarity before modeling the model ARIMA(p,d,q); if not, differencing is needed.

C. XGBOOST

Boosting methods in ensemble learning are based on gradient lifting, a technique for optimizing the model. Ensemble learning is a popular area of research in machine learning that tries to develop a powerful model for precise predictions by combining several weak learners. Ensemble learning uses an aggregated model that trains the best combination of weak learners to produce more accurate outcomes than a single model. XGBoost is an integrated learning algorithm that uses parallel decision trees to produce collective decisions. Each decision tree's input is linked to the prior tree's training and prediction outcomes, resulting in an increasingly accurate decision function with each iteration. XGBoost is one of the ensembles learning techniques used for prediction modeling. The prediction of the XGBoost model can be written as one of the ensembles learning techniques as follows:

$$y = \sum_{i=1}^k f_k(x_i), \quad (8)$$

where k represents the k th tree, shows the total quantity of trees, as well indicates the accuracy of the sample's prediction. The following is a description of the objective function:

$$\text{Obj}(\theta) = \sum_{i=1}^n l(y_i, y_i) + \sum_{k=1}^k \Omega(f_k), \quad (9)$$

where $l(y_i, y_i)$ is the training error of the sample, $\sum_{k=1}^k \Omega(f_k)$ denotes the k -th tree's regular term that keeps the model from being overfit.

D. EVALUATION

The three different measures of prediction error used in the study to evaluate metrics are root mean square error (RMSE), mean absolute error (MAPE), mean absolute percent error (MAPE) and mean absolute scaled error (MASE). While RMSE and MAE show the size of the error, MAPE represents the degree of inaccuracy as a percentage. MASE is especially helpful when working with time series data or comparing the performance of several models. To assess the model from a variety of angles by combining MAPE, RMSE, and MAE. MAE is robust to outliers, RMSE is sensitive to big errors, and MAPE is sensitive to tiny values. When combined, they offer a more comprehensive and equitable evaluation of model performance. All three criteria have also been established by Hansun et al. [36], [38] and Shahid et al. [37].

$$\text{MAE} = \frac{1}{o} \sum_{l=1}^o |Y_l - F_l|, \quad (10)$$

$$\text{RMSE} = \sqrt{\frac{1}{o} \sum_{l=1}^o (Y_l - F_l)^2}, \quad (11)$$

$$\text{MAPE} = \frac{1}{o} \sum_{l=1}^o \frac{|Y_l - F_l|}{Y_l},$$

$$\text{MASE} = \frac{1}{n} \sum_{t=1}^n \frac{y_t - \hat{y}_t}{\text{MAE}}, \quad (12)$$

where n overall value set, the actual value is shown through Y_l , and F_l denotes as the anticipated amount. Another study evaluates the matrices using The Diebold_Mariano (DM). the statistic can be utilized in comparison to the forecasting of the two model's performance, as described in [39] and [40]. The formula for the DM statistic is given by Equation (13):

$$\text{DM} = \frac{\bar{k}}{\sqrt{\text{Var}(\bar{k})}}, \quad (13)$$

where $\bar{k} = \frac{1}{l} \sum_{l=1}^o k_l$, $k_l = (Y_l - F_{1,l})^2 - (Y_l - F_{2,l})^2$, $\text{Var}(\bar{k}) = \frac{1}{l}(z_0 + 2 \sum_{o=1}^k z_o)$ and $z_o = \text{Cov}(k_l, k_{l-o})$.

where $\bar{k}$ is the mean of the squared forecast errors (k_l), which shows the calculated differential between the real and expected values (Y_l) from each model ($F_{1,l}$ and $F_{2,l}$). The variance of $\bar{k}$ is calculated using Equation (13) with t being the length of the forecast period, z_0 being the variance of the forecast errors squared, and z_o being the covariance between the squared forecast errors at both times l and $l-o$. The first and second models' projected values are shown by $F_{1,l}$ and $F_{2,l}$, respectively.

The Directional Statistic (DS) gauges the proportion of accurate directional signals generated by a forecasting model and serves as an indicator for directional forecast accuracy. To calculate DS, one must add up all of the correct directional signals that a model has generated over a given period and express this as a percentage of the total number of signals that have been anticipated.

$$\text{DM} = \frac{1}{m} \sum_{l=1}^m F_l \times 100, \quad (14)$$

here $F_l = 1$, if $(Y_{l+1} - Y_l) \times (\hat{Y}_{l+1} - Y_l) \geq 0$ otherwise $F_l = 0$

Equation (14) provides the formula for computing DS, where 'm' is the total number of directional forecasts, $X(l)$ is the real value of the variable being forecasted at time l , and Y_l is the value that is expected at that moment. F_l takes a value of 1 or 0, based on whether the directional signal produced by the model is accurate or inaccurate.

III. THE EMPIRICAL ANALYSIS

A. STATISTICAL OVERVIEW OF THE DATA

The US Energy Information Administration (EIA) provided information regarding WTI (West Texas Intermediate) crude oil for this study, which was divided into 75/25 training and testing sets for forecasting accuracy between March 12, 2018, and February 14, 2022. The testing set comprised 250 samples from February 24, 2021, to February 14, 2022, while the training set had 749 samples between February 23, 2021, and March 12, 2018. Figure 2 depicts a sharp decline in WTI prices until January 11, 2020, then another decline until May 11, 2021. The period from May 12, 2021, to February 14, 2022, saw a clear increasing trend. The data is normal, according to the Jarque-Bera (J-B) test, however, Table 1's PP and ADF statistics indicate that the sequence is not stationary. For clearer picture the data is presented in Figure 2.

B. DATA DECOMPOSITION AND RECONSTRUCTION

This research aims to propose a method for reconstructing PFs derived from LMD. The PFs are split into two categories: deterministic and stochastic, which are then modeled separately using different techniques, namely ARIMA and XGBOOST models. The ultimate forecasting is then performed using the combined individual deterministic PFs. The WTI price data is decomposed by LMD with a maximum of 20 generating function components and 30 iterations. The entire framework of the suggested method is depicted in Figure 1. The decomposition's findings reveal that Three PFs plus a residual make up the WTI CO price sequence.

According to Figure 3, the CO price sequence can be divided into a residual series and three PFs., with the latter showing a downward trend. Based on the mutual information of each component, the PFs are reconstructed, and the AMI plots for the PFs are visually evaluated. It has been noted that the remaining PFs are deterministic, but the initial PF is stochastic. The second



FIGURE 2. WTI market (Dollar per Barrel) daily crude oil prices.

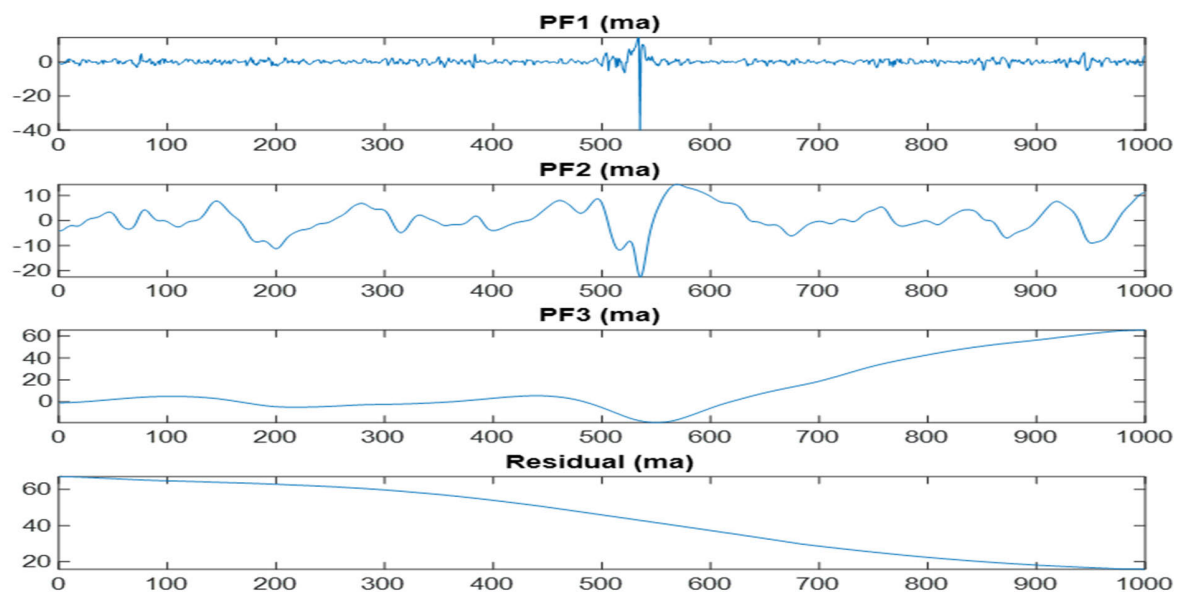


FIGURE 3. Using "LMD" the PFs and residual plots of the WTI crude oil price".

through fourth PFs are added as deterministic components, and the first PF is treated as stochastic, resulting in two components. All of the deterministic series and both stochastic and deterministic graphs are added to produce the virtual product functions (VPF). are shown in Figures 4 and 5 respectively.

Particularly, Deterministic factors include PF2, PF3, and PF4, PF1 is seen as stochastic. To construct SD components, these components are maintained separately and together. The time series model's stationarity is the fundamental presumption for applying the ARMA model. To make time-series stationary, successive differences are taken. [41], [42]. To ascertain whether the time series is stationary, perform the ADF test. [43]. After getting a stationary time series, choose

the correct time series model by correctly ordering the AR and MA terms. The best ACF and PACF graphs in order are used for selection. The Model uses the Adam algorithm for cross-validation and training. Accordingly, a 75:25 data ratio is set aside for testing and training. The model's appropriateness is assessed using the Ljung-Box (LB) test. For WTI CO prices based on ACF and PACF plots the ARIMA(2,1,3) model was chosen. By modifying the ARIMA model for training and testing, forecasts are made. We fit the data with the ARIMA(2,1,3) model using R. The standardized residuals, ACF and lag plots of the WTI ARIMA fitted model are shown in Figure 6.

The time series model must be steady to use the ARMA model, which can be accomplished by taking successive

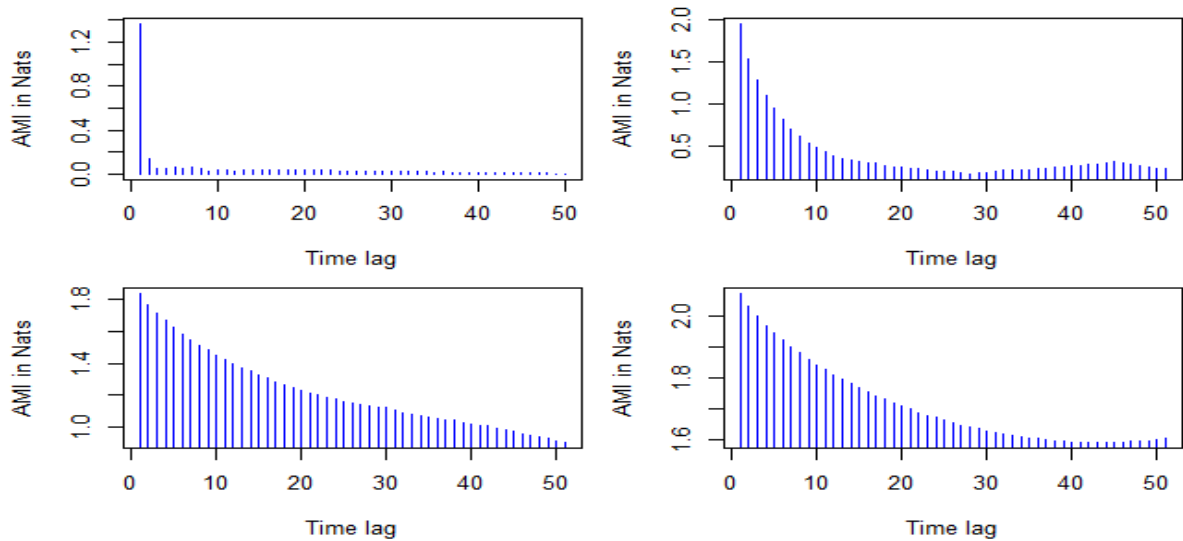


FIGURE 4. “PF’s plots for Average Mutual Information (AMI).

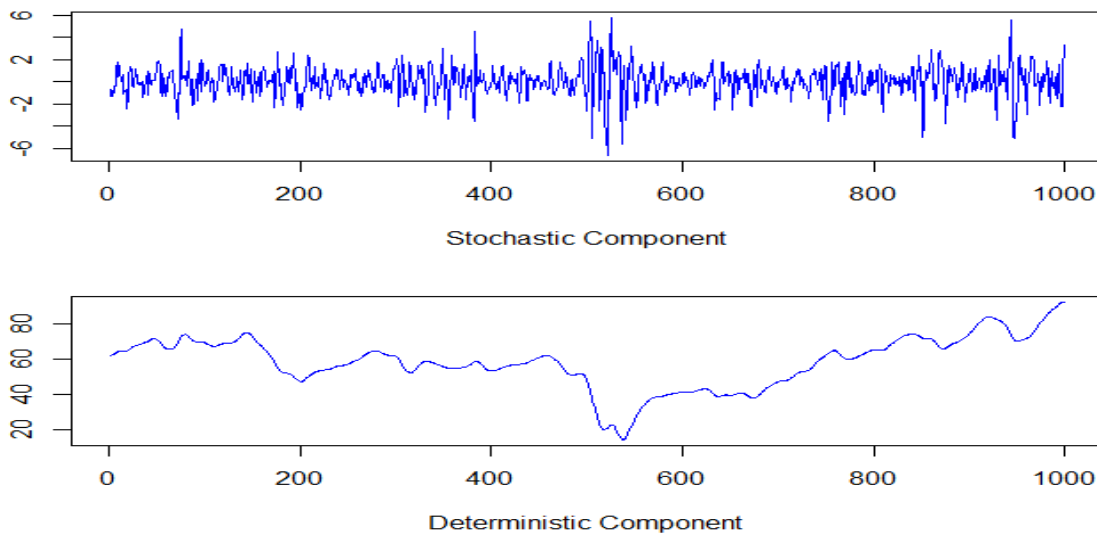


FIGURE 5. Plots of deterministic and stochastic components.

differences. The time series’ stationarity is assessed using the ADF test. The AR and MA terms are picked based on the best-order ACF and PACF plots to choose the right model for the time series. With a data ratio of 75:25 selected for instruction and assessment, the Adam algorithm is employed for training and cross-validation. The Ljung-Box test is employed to determine if the model is adequate. Using R software, the ARIMA model is fitted for the forecasting training and testing periods. Figure 6 shows the residual plot of the WTI ARIMA fitted model with ACF lag.

Based on the ADF and PP tests, it is determined that PF1 and VPF are non-stationary, demonstrating the necessity of differencing. For the five-day-ahead time horizon, both PF and VPF have a difference order of 1, whereas VPF has a difference order of 2 for the ten-day-ahead and

fifteen-day-ahead models. PF and VPF are fitted for PF1 and VPF in the LMD-ARIMA model, which stand in for the stochastic and deterministic components, respectively, which are fitted and added to the LMD-SD-ARIMA model.

C. XGBOOST

For prediction purposes, the residuals from LMD-SD-ARIMA and “LMD-ARIMA” are used in a hybrid XGBOOST model. The baseline XGBoost algorithm’s parameter settings are as follows: $\text{max_depth} = 7$, $\text{learning_rate} = 0.3$, and $\text{N_estimators} = 30$. Out of all the prediction models, XGBoost has the lowest errors. The XGBoost model theoretically explains this by applying the second-order Taylor expansion to the loss function in the model training and makes use of the first-order and

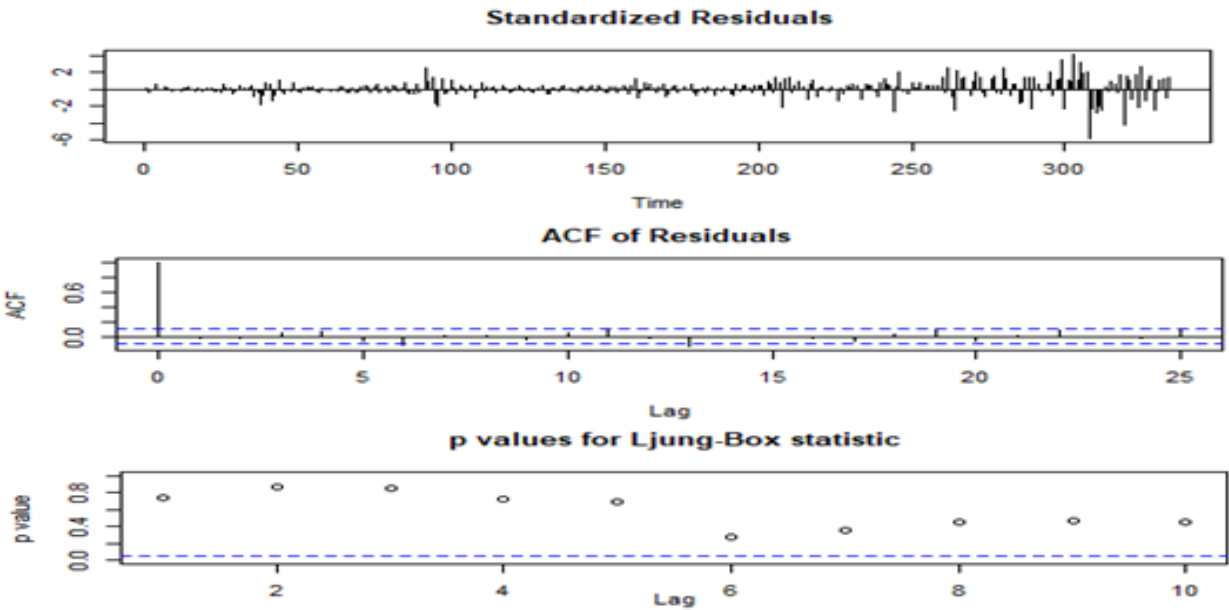


FIGURE 6. ARIMA fitted model (WTI).

TABLE 1. Values for the ADF test and the PP test (WTI, all PFs, and VPF).

Time Horizon	PF1		VPF	
	ADF	PP	ADF	PP
5 days ahead	-4.290 (0.500)*	4.200 (0.990)*	-1.570 (0.760)*	-7.346 (0.700)*
10 days ahead	-4.167 (0.510)*	4.255 (0.990)*	-1.616 (0.740)*	-7.324 (0.700)*
15 days ahead	-4.400 (0.530)*	4.306 (0.990)*	-1.604 (0.750)*	-7.296 (0.700)*

Note: * indicates rejections of the null hypothesis, which is non-stationary, at the 5% significance level.

second-order derivative information of the training samples to optimize the model structure, ensuring that the model training is adequate. However, a greater forecasting inaccuracy results from XGBoost’s estimated load curve trailing behind the actual load curve. To increase its accuracy, further XGBoost model-based optimization is therefore required.

The LMD-ARIMA-SD-XGBOOST hybrid prediction model improves forecasting accuracy and fixes the lag issue when compared to the XGBoost model. The enhancement illustrates the necessities of series breakdown. In multiple scenarios using various industrial customer datasets, LMD-ARIMA-SD-XGBOOST outperforms LMD-XGBOOST in terms of stability and error reduction. The LMD method’s superiority over the EEMD approach in terms of series decomposition can be attributed to its ability to overcome modal aliasing faults and boundary effects.

IV. OUTCOMES AND DISCUSSIONS

A new forecasting hybrid design called LMD-SD-ARIMA-XGBOOST was proposed and compared with four other methods: ARIMA, XGBOOST, LMD-ARIMA, and LMD-ARIMA-XGBOOST. The efficiency of the models was assessed by computing RMSE, MAE, and MAPE using WTI data. Table 5 gives an overview of the accuracy of each model’s performance.

The LMD-SD-ARIMA-XGBOOST hybrid model had the model with the lowest RMSE overall, at 0.1504, enabling the use of this approach to forecast the price of crude oil. Secondly, regarding the traditional econometric model, Accuracy assessments of each model (ARIMA, for example) and decomposition-ensemble models (LMD-ARIMA, LMD-ARIMA-XGBOOST, etc.) were compared with one another, and the latter’s accuracy in forecasting was shown to be lower than the former. However,

TABLE 2. Measures of all VPS, PFs, and “WTI” crude oil prices’ accuracy.

ARIMA	(p,d,q)	MAE	MAPE	RMSE	MASE	AIC	L-B
Five-day-ahead selected model results							
PF	(1,1,1)	0.0004	0.0008	0.0017	0.85	-11180.71	0.2705(0.6030)*
VPF	(2,1,2)	0.6909	74.9686	0.8944	0.89	2345.21	0.4873(0.4851)*
Ten-day-ahead selected model results							
PF 1	(1,1,1)	0.0008	0.1310	0.0010	0.86	-9733.25	0.7977(0.3718)*
VPF	(1,2,2)	0.0002	0.0033	0.0009	0.86	-10268.45	5.2575(0.0219)*
Fifteen-day-ahead selected model results							
PF1	(2,1,1)	0.0007	0.1306	0.0010	0.91	-9683.59	0.8381(0.3600)*
VPF	(1,2,2)	0.0002	0.0033	0.0009	0.89	-10205.47	5.2258(0.0223)*

TABLE 3. Accuracy of prediction for “WTI”.

Method	MAE	MAPE	RMSE	DS
A) Five-day-ahead select model results.				
ARIMA	2.02	0.337	2.1973	66.23
XGBOOST	7.4865	0.125	8.0739	69.15
LMD-ARIMA	1.5188	0.0254	1.5917	53.63
LMD-XGBOOST	11.9011	0.1983	13.9352	59.12
LMD-SD-XGBOOST	21.2708	0.3567	21.2954	69.24
LMD-ARIMA-SD-XGBOOST	1.3341	0.0241	1.4842	72.34
B) Ten-day-ahead select model results.				
ARIMA	1.5154	0.026	1.6229	66.67
XGBOOST	8.4678	0.1462	9.7839	69.56
LMD-ARIMA	1.3246	0.0229	1.4936	60.23
LMD-XGBOOST	20.509	0.3543	24.9412	63.12
LMD-SD-XGBOOST	19.879	0.3396	19.9542	69.54
LMD-ARIMA-SD-XGBOOST	1.1587	0.02	1.3881	75.56
C) Fifteen-day-ahead select model results.				
ARIMA	1.1286	0.0192	1.3483	61.35
XGBOOST	8.6651	0.1482	9.7931	63.35
LMD-ARIMA	1.1785	0.0199	1.4921	65.56
LMD-XGBOOST	19.4858	0.3337	22.8925	70.12
LMD-SD-XGBOOST	19.9708	0.3417	20.0235	71.34
LMD-ARIMA-SD-XGBOOST	0.9464	0.0161	1.1231	76.67

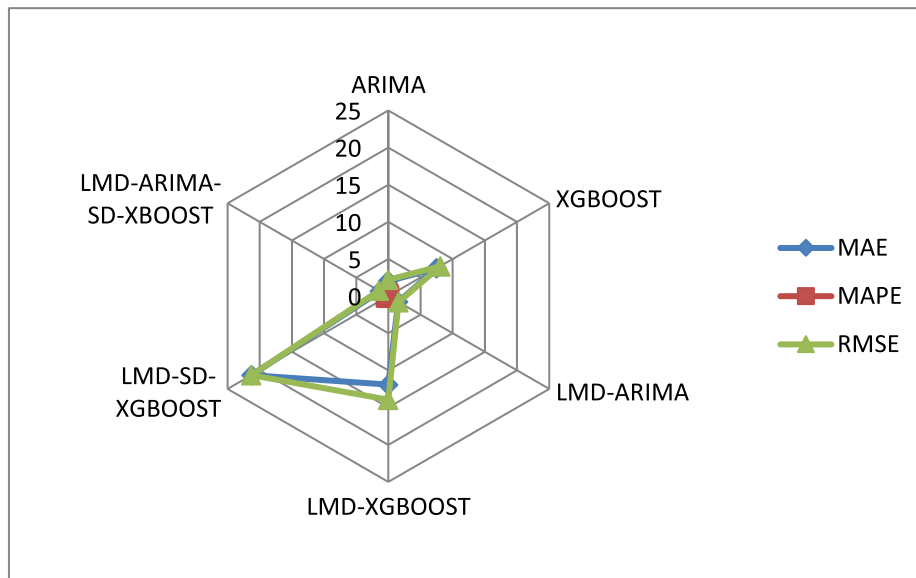


FIGURE 7. "Accuracy measurements" comparing other models with the "hybrid" model (five days ahead).

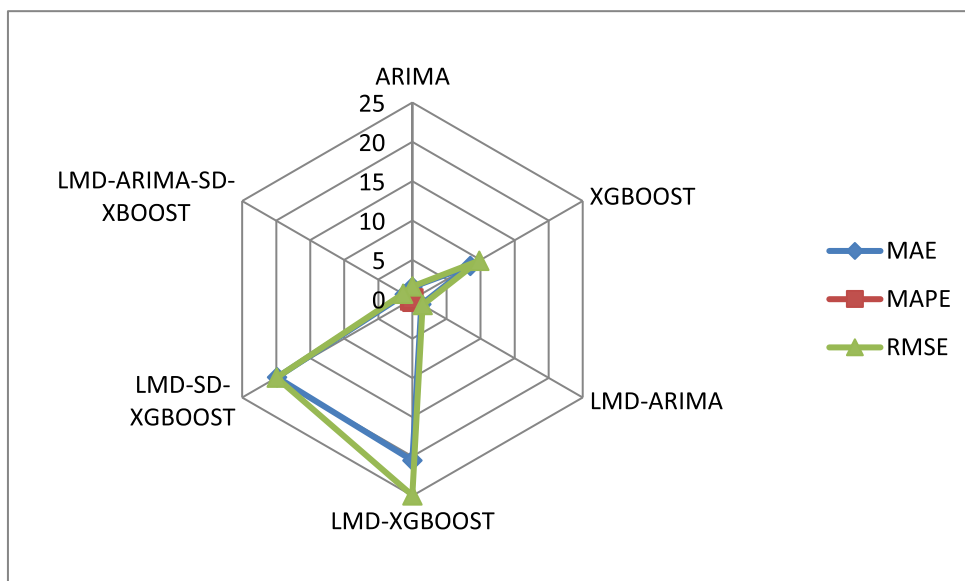


FIGURE 8. "Accuracy measurements" comparing other models with the "hybrid" model (ten days ahead).

when a single model's three accuracy metrics (XGBOOST) and the (LMD-ARIMA-XGBOOST and LMD-SD-ARIMA-XGBOOST) decomposition-ensemble models are contrasted for a machine learning model. The decomposition-ensemble models' accuracy metrics are not as good as those of a single model. It demonstrates how the accuracy of the traditional econometric model used to predict oil prices can be enhanced following decomposition and reconstruction. The LMD-SD-ARIMA-XGBOOST framework can easily be modified to predict variables in non-energy sectors or to forecast prices in other volatile markets, such as natural gas or precious metals.

Future research directions are briefly outlined in the conversation. These could involve combining complementary machine learning approaches, investigating possible improvements to improve performance in various contexts, or testing the model with a variety of datasets to confirm its resilience and generalizability. This might pave the way for further advancements in hybrid forecasting techniques and wider implementation by doing this.

A. FORECASTING RESULTS

Table 4 compares 20 days starting on February 15, 2022, and ending on March 15, 2022, at the original WTI oil

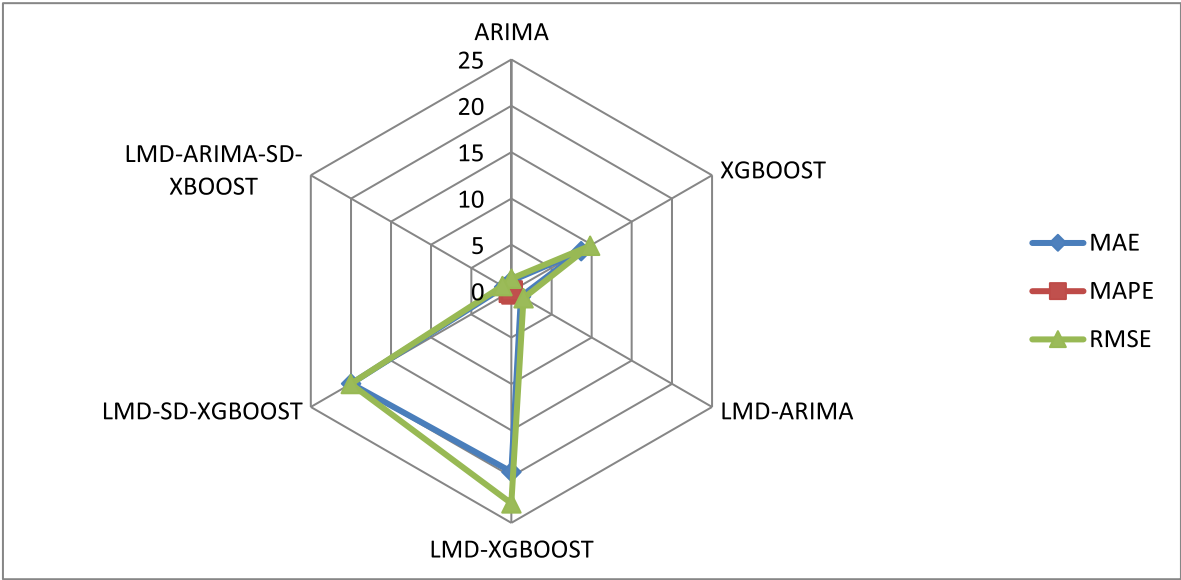


FIGURE 9. Accuracy measurements” comparing other models with the “hybrid” model (fifteen days ahead).

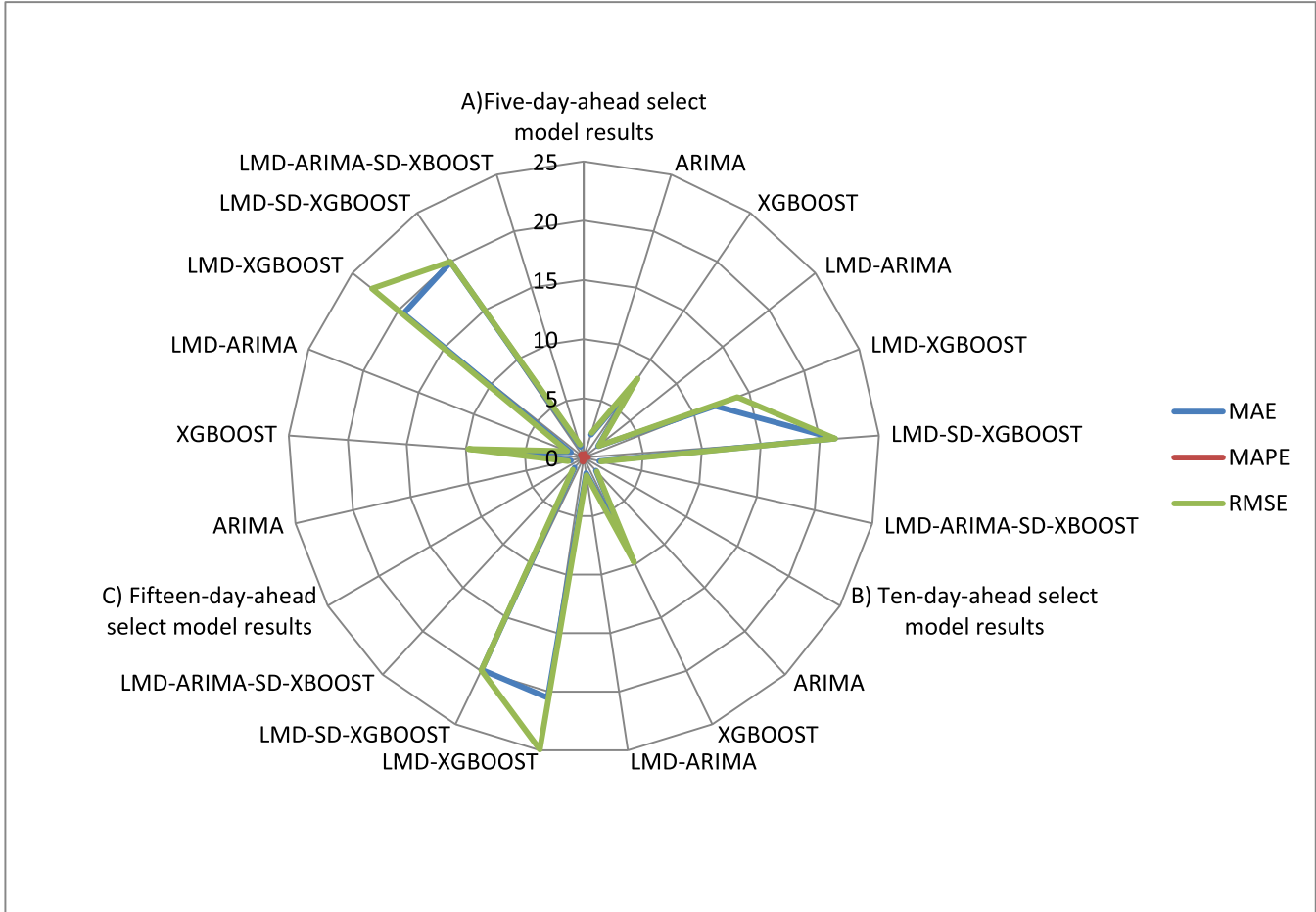


FIGURE 10. Accuracy measurements for 5, 10 and 15 days ahead.

TABLE 4. WTI forecasting results using ARIMA, XGBOOST, LMD-ARIMA, LMD- XGBOOST, LMD-SD- XGBOOST, and LMD -ARIMA-SD- XGBOOST models.

Date	WTI	ARIMA	XGBOOST	LMD- ARIMA	LMD- XGBOOST	LMD-SD - XGBOOST	LMD-ARIMA-SD- XGBOOST
2/15/2022	92.07	84.709	83.216	84.709	87.751	88.423	90.138
2/16/2022	93.66	85.073	83.307	85.073	87.67	88.698	91.414
2/17/2022	91.76	85.719	83.139	85.719	87.77	88.873	89.585
2/18/2022	91.07	85.999	82.694	85.999	87.718	88.948	89.656
2/22/2022	92.35	86.471	82.006	86.471	87.781	89.935	90.645
2/23/2022	92.1	86.722	81.138	86.722	87.747	89.849	90.567
2/24/2022	92.81	87.06	80.195	87.06	87.786	89.705	90.437
2/25/2022	91.59	87.248	79.208	87.248	87.765	88.511	89.265
2/28/2022	95.72	87.461	78.16	87.461	87.788	90.283	92.064
3/01/2022	103.41	107.57	77.094	107.57	107.775	95.031	97.105
3/02/2022	110.6	107.671	106.051	107.671	107.7101	105.710	106.621
3/03/2022	107.67	107.716	105.015	107.716	107.7102	105.513	106.395
3/04/2022	115.68	107.749	103.995	107.749	107.7102	105.243	106.16
3/07/2022	119.4	107.739	103.046	107.739	107.7106	104.969	105.919
3/08/2022	123.7	107.726	102.172	107.726	107.7102	104.691	105.673
3/09/2022	108.7	107.692	101.312	107.692	107.7101	104.413	105.424
3/10/2022	106.02	107.661	100.451	107.661	107.7102	104.144	105.110
3/11/2022	109.33	107.622	109.637	107.622	107.7109	103.101	104.947
3/14/2022	103.01	107.510	101.107	107.510	107.710	103.632	104.714
3/15/2022	96.44	87.556	88.21	87.556	87.781	83.382	84.482

prices with 20 days of forecasted outcomes for all six models.

Figures 10 show a visual comparison between the outcomes of oil price estimates using all six models and compares the proposed model’s mistakes to those of the other models. The three figures demonstrate that

LMD-SD-ARIMA-XGBOOST has the best forecasting performance. The “LMD-SD-ARIMA-XGBOOST” approach also does better in the long-term projection than in the short- and medium-term projection. Additionally, Table 4 demonstrates that there are considerable differences across all of the models.

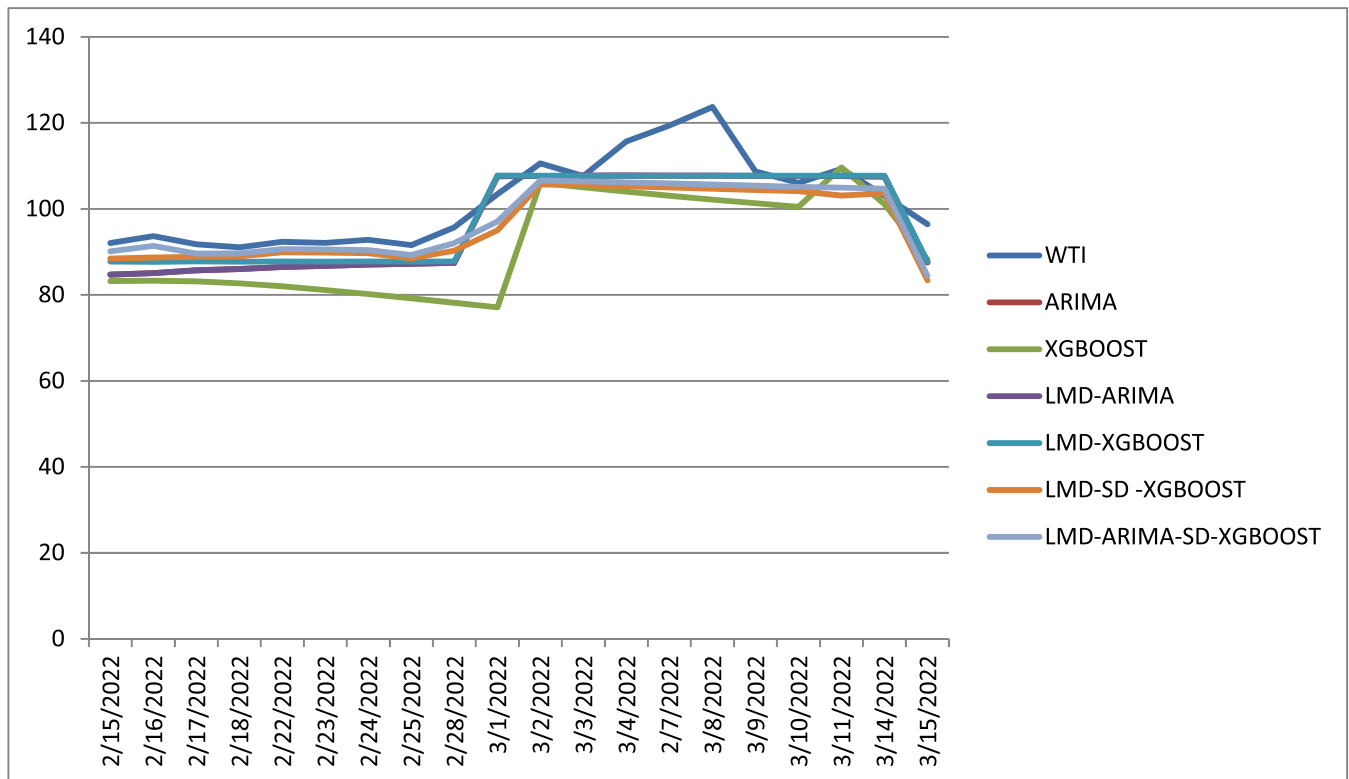


FIGURE 11. Forecasting results using ARIMA, XGBOOST, LMD-ARIMA, LMD- XGBOOST, LMD-SD- XGBOOST, and LMD-ARIMA-SD-XGBOOST models.

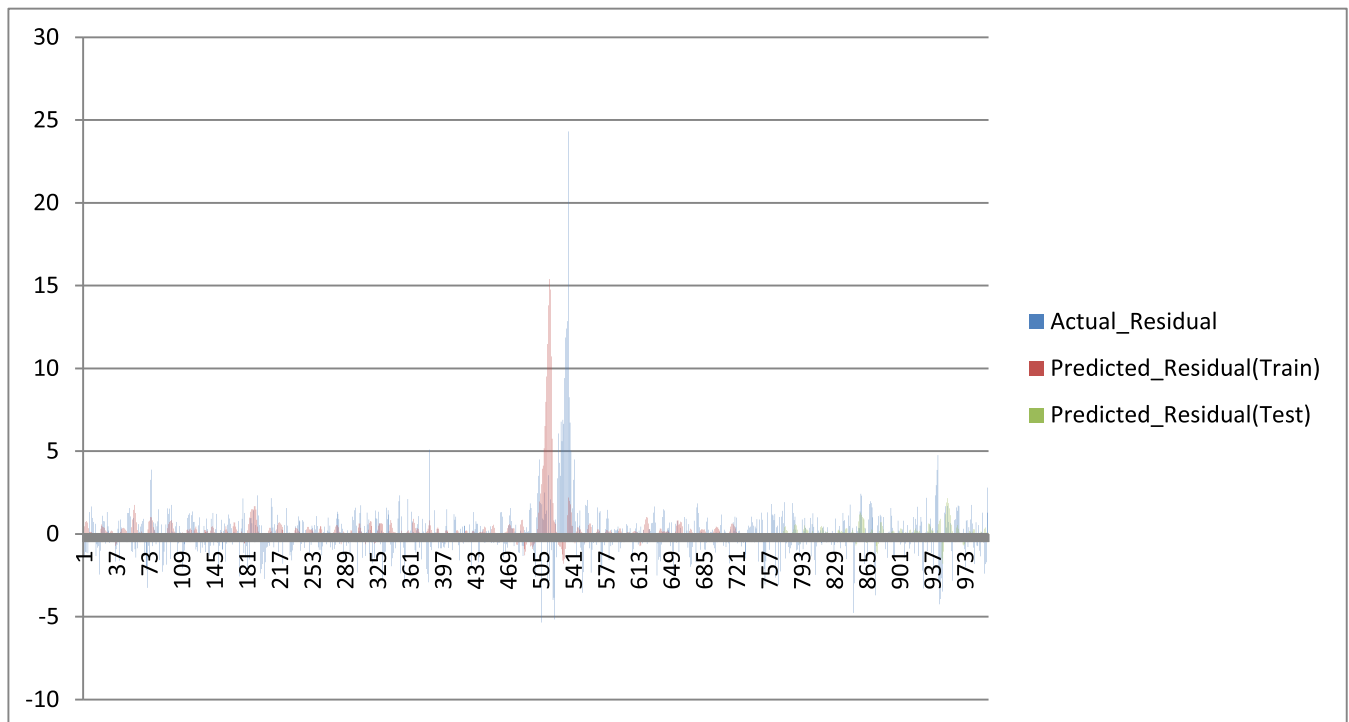


FIGURE 12. "LMD-SD-ARIMA" (five days ahead) residual_model fitting using XGBOOST model.

V. CONCLUSION

This study presents LMD-SD-ARIMA-XGBOOST, a novel hybrid forecasting approach created to address the chal-

lenges of predicting the price of crude oil (WTI CO). The method surpasses five current approaches and establishes a new standard for forecast accuracy by separating deter-

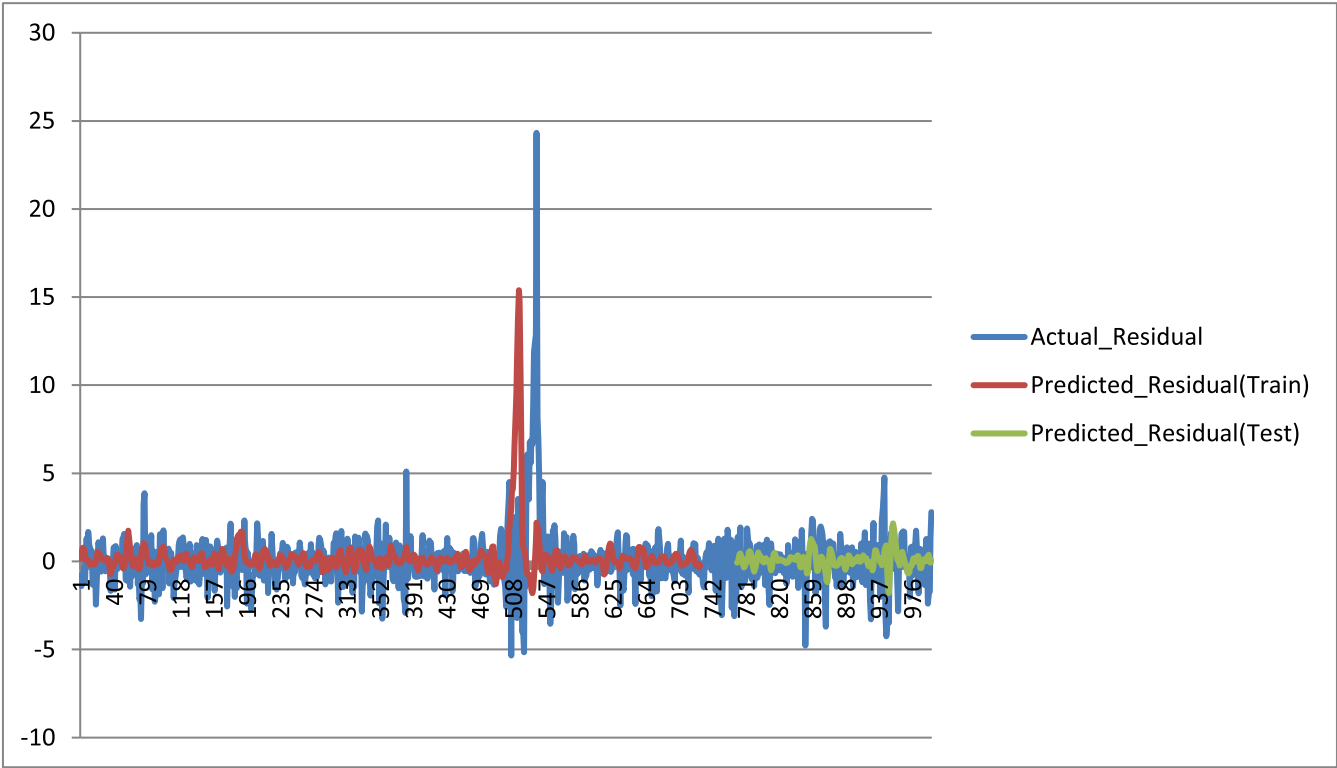


FIGURE 13. “LMD-SD-ARIMA” (ten days ahead) residual_model fitting using XGBOOST model.

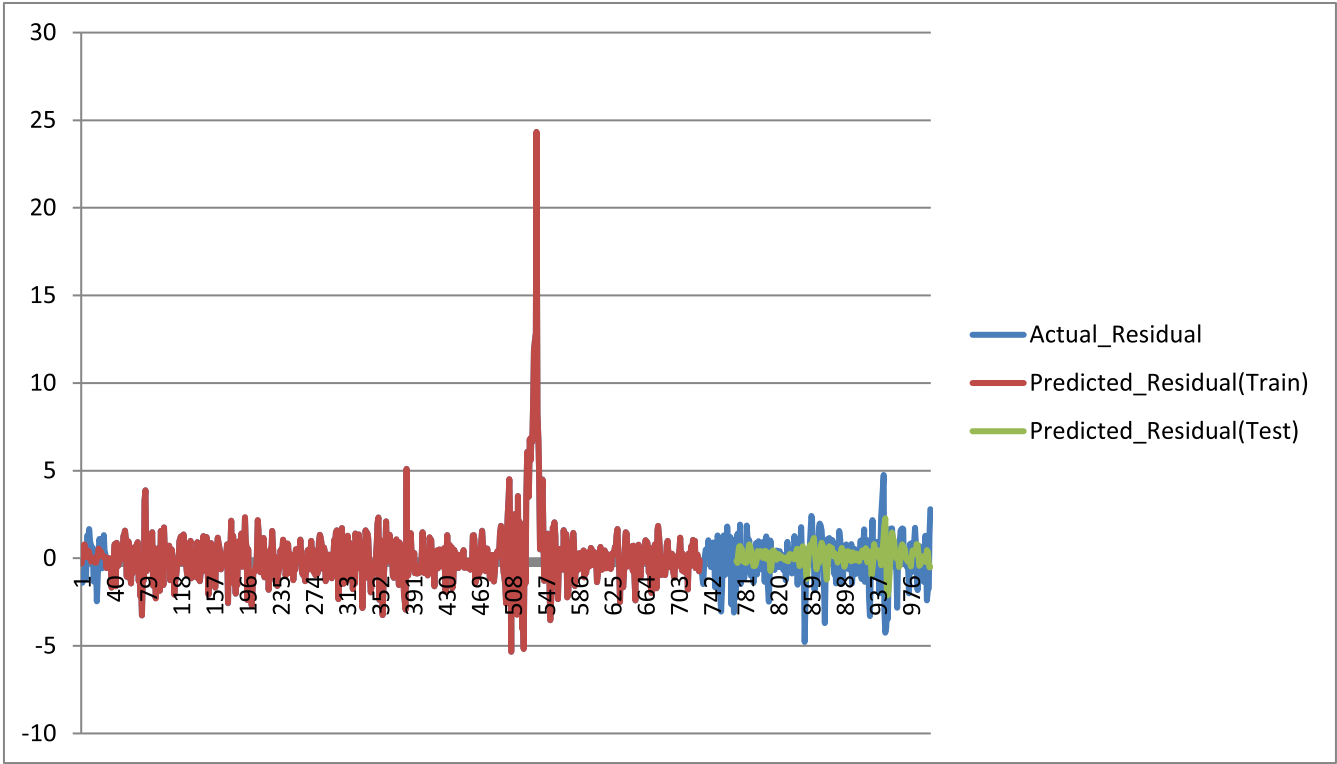


FIGURE 14. “LMD-SD-ARIMA” (fifteen days ahead) residual model fitting using XGBOOST model.

ministic and stochastic components from stationary and non-stationary PFs. Key findings demonstrates that decomposition and reconstruction greatly improve the accuracy of

long- and medium-term projections when combined with sophisticated hybrid models. This improvement provides a more sophisticated view of market dynamics by going beyond

TABLE 5. Diebold mariano test for all selected models.

Models	Comparison	DM statistic	p-values
XGBOOST	ARIMA	-4.323	1.26E-04
LMD-ARIMA	ARIMA	0.0	3.35E+04
LMD-SD-ARIMA	ARIMA	3.464	8.41E-05
LMD-ARIMA- XGBOOST	ARIMA	-2.264	2.35E-03
LMD-ARIMA-SD- XGBOOST	ARIMA	-1.253	2.22E-02
LMD-ARIMA	XGBOOST	4.636	2.64E-08
LMD-SD-ARIMA	XGBOOST	4.864	3.64E-06
LMD-ARIMA- XGBOOST	XGBOOST	5.6473	1.54E-8
LMD-ARIMA-SD- XGBOOST	XGBOOST	5.4324	3.42E-6
LMD-SD-ARIMA	LMD-ARIMA	3.455	7.42E-07
LMD-ARIMA- XGBOOST	LMD-ARIMA	-2.543	2.26E-03
LMD-ARIMA-SD- XGBOOST	LMD-SD-ARIMA	-2.043	2.65E-05
LMD-ARIMA- XGBOOST	LMD-SD-ARIMA	-2.432	4.26E-03
LMD-ARIMA-SD- XGBOOST	LMD-SD-ARIMA	-1.543	6.56E-05
LMD-ARIMA-SD- XGBOOST	LMD-ARIMA-XGBOOST	5.675	1.54E-06

conventional econometric models. The finding has significant implications: precise oil price forecasting becomes a possibility, enabling business leaders and politicians to react to market swings in advance. This creative strategy could promote economic stability and strengthen the energy sector's resilience in a global market that is becoming more and more turbulent by offering a strong foundation for strategic decision-making.

A. FUTURE RECOMMENDATIONS

Here machine learning techniques can also be employed for comparative analysis with the proposed hybrid EEMD-XGBOOST model. Furthermore, the impact of exogenous variables on the forecasting accuracy can also be investigated.

Applying the suggested model to various economic and financial time series, including commodity, stock, and currency rates, is another extension that could be made. For these datasets, it would be interesting to contrast the hybrid model's performance to that of alternative models that are already known to exist in the literature.

In addition, the interpretability of the model can also be improved by using feature importance analysis and partial dependence plots. This will provide insights into the most important variables and their impact on the forecasting results.

Lastly, the proposed model can be extended for online forecasting, where new data are continuously added to the dataset. This will allow us to provide real-time forecasts and adjust the model parameters accordingly.

LIST OF ACRONYMS

ARIMA	Auto regressive moving average
DM	Diebold-Mariano
EMD	Empirical Mode Decomposition
EEMD	Ensemble empirical mode Decomposition
MAPE	Mean Absolute Percentage Error
MAE	Mean Absolute Error
IMF	Intrinsic Mode Function
PF	Product Function
XGBOOST	Extreme Gradient Boosting technique
MSE	Mean square Error

MASE Mean Absolute Scaled Error
DS Directional Statistics

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